

UNIVERSITY OF WAH(UOW)
DEPARTMENT OF COMPUTER
SCIENCE



Lab Task # 09

COURSE: Data Structure & Algorithm Lab_201L

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Task # 01

```
using namespace std;
int main() {
    stack<string> history;
    history.push("Page1");
    history.push("Page2");
    history.push("Page3");
    history.push("Page4");
    history.push("Page5");
    history.pop();
    history.pop();
    cout << "Current Page: " << history.top() << endl;
    cout << "Pages remaining in history: " << history.size() << endl;
    return 0;
}
```

Output

```
Current Page: Page3
Pages remaining in history: 3
```

Task # 02

```
#include <iostream>
using namespace std;
const int MAX 5;
int main() {
    int stack[MAX];
    int top = -1;
    stack[++top] = 10;
    stack[++top] = 20;
    stack[++top] = 30;
    top--;
    top--;
    cout << "Top element: " << stack[top] << endl;
    cout << "Total elements in stack: " << top + 1 << endl;

    return 0;
}
```

Output

```
Top element: 10
Total elements in stack: 1
```

Task # 03

```
struct Node {
    int data;
    Node* next;
};
Node* top = NULL;

void push(int actionID) {
    Node* newNode = new Node();
    newNode->data = actionID;
    newNode->next = top;
    top = newNode;
}

void pop() {
    if (top == NULL) {
        cout << "No actions to undo" << endl;
        return;
    }
    Node* temp = top;
    top = top->next;
    delete temp;
}
```

```
void displayTop() {
    if (top != NULL)
        cout << "Most recent active action: " << top->data << endl;
    else
        cout << "No active actions" << endl;
}

void isEmpty() {
    if (top == NULL)
        cout << "Stack is empty" << endl;
    else
        cout << "Stack is not empty" << endl;
}
```

```
int main() {  
  
    push(101);  
    push(102);  
    push(103);  
    push(104);  
    push(105);  
  
    pop();  
    pop();  
    pop();  
  
    displayTop();  
  
    isEmpty();  
    pop();  
    pop();  
    pop();  
    return 0;  
}
```

Output

```
Most recent active action: 102  
Stack is not empty  
No actions to undo
```